

Yiting Liang

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EDUCATION

The University of New South Wales (UNSW)

Master of Engineering Science (Robotics Specialisation)

Weighted Average Mark (WAM): 88.636/100 (Good Standing Academic Status)

Sydney, Australia

09/2024- 09/2026 (expected)

Capital Normal University

Bachelor of Engineering in Artificial Intelligence

Overall GPA: 3.19/5

Beijing, China

09/2020- 06/2024

RESEARCH INTERESTS

Autonomous Robotics, Robot Perception with Computer Vision & Deep Learning, Human-Robot Interaction

Awards & Honors

Third Prize, University-Level Programming Competition, Capital Normal University, Sep 2021;

Third-Class Scholarship, Capital Normal University, Dec 2023

Core Courses

Computer Vision (87/100, High Distinction, HD); Mechanics of Solids (97/100, HD); Feedback Control Systems (92/100, HD); Robot Design (84/100, Distinction); Control of Robotic Systems (In Progress)

Operating Systems; Electronic Technology Practice; Comprehensive Practice of AI Systems; Natural Language Processing; Machine Learning; Intelligent Robot Technology and Simulation

RESEARCH EXPERIENCE

Quadcopter Charging/Docking Station with Auto Landing

2025.09–present

Supervisor: Dr. Mohammad Deghat, UNSW

- Designed a low-cost, infrastructure-independent autonomous landing system for quadcopters
- Built a modular development pipeline using ROS 2 Humble and Gazebo Harmonic, and evaluated system performance using PID and SMC-PID controllers under disturbance-rich scenarios
- Investigated robust control and dynamic docking under realistic disturbances, strengthening research interest in UAV autonomy and resilience in extreme operational conditions
- Achieved high-precision autonomous landing under ground effect, with position error within 10 mm, validated in a motion capture system

PROJECT EXPERIENCE

Vision-Guided 6-DOF Manipulator Handling System

2025.06–2025.08

Supervisor: Associate Professor Hoang-Phuong Phan, UNSW; MTRN4230 Project

- Developed a vision-guided pick-and-place system for 6-DOF robotic arm with vacuum gripper, realizing workpiece transportation and obstacle avoidance.
- Completed coordinate transformation via homography and ArUco markers; implemented Artificial Potential Field (APF) for path planning in MATLAB.
- Built custom inverse kinematics solver, controlled the manipulator via RTDE to finish grasping and precise placement at fixed height.

Micro-mouse Maze-Navigation Robot

2025.06–2025.08

Supervisor: Dr. Will Midgley, UNSW; MTRN3100 Group Project

- Built a robot capable of navigating a maze in the shortest time using autonomous mapping and path-planning
- Contributed for autonomous mapping and path-planning modules, implementing Flood Fill and BFS algorithms to optimize route efficiency (reduced navigation time in test mazes)
- Tuned PID controllers for motor speed and direction, improving the robot's stability on uneven maze surfaces
- Conducted hardware assembly and debugging, ensuring the robot completed maze runs without mechanical failures

Computer Vision for Remote Sensing Image Classification

2025.02–2025.05

Supervisor: Prof. Yang Song, UNSW; COMP9517 Group Project

- Compared traditional machine learning and deep learning methods for aerial scene classification (15 categories)

with applications in urban planning

- Implemented a Random Forest classifier using handcrafted features (LBP, HOG, GLCM), achieving 72% accuracy on the test dataset
- Fine-tuned pre-trained deep learning models including ResNet18 and EfficientNet with data augmentation, boosting accuracy to 89%, outperforming traditional methods by 17%
- Co-authored the final report, analyzing trade-offs between model complexity and real-time applicability for robotic aerial surveillance

Graph Neural Networks (GNN) for RGB-D Semantic Segmentation

2023.04–2023.07

Supervisor: Prof. Na Guan; Undergraduate Group Project

- Applied GNN to 3D point cloud semantic segmentation using RGB-D data, with potential for robotic perception.
- Converted RGB-D data to HHA-encoded 3D point clouds and used KNN to construct graph isomorphism for GNN input

INTERNSHIP EXPERIENCE

Beijing Tianyu Zhiyuan Technology Development Co., Ltd.

2024.06–2024.08

PLC Programming Intern, SOE Function Module Development

- Developed a software-based Sequence of Events (SOE) module for Rockwell Logix5000 PLC to achieve millisecond-level event logging without additional hardware costs.
- Programmed and deployed dedicated logic inside PLC for real-time event capture, timestamping and in-memory storage of alarms, faults and equipment shutdown signals.
- Built corresponding HMI upper computer programs to extract, query and migrate SOE data from PLC memory to local databases for subsequent analysis.
- Designed robust fault-tolerance mechanisms to ensure continuous data recording during communication interruptions or HMI failures, enabling automatic resume of data transmission once connections are restored.
- Mastered Logix5000 programming tools and completed full-cycle coding, commissioning and functional verification of the data transmission module.

Python Algorithm Development Intern, Automotive PBS Conveyor Line Intelligent Sorting System

- Designed and implemented intelligent path scheduling algorithms for automotive PBS conveyor lines to reorder car bodies and match the production sequence of final assembly workshops.
- Coded, debugged and encapsulated the core routing algorithm with Python, providing reusable functional modules for the central control system.
- Realized seamless collaboration between algorithm modules and field PLC systems to deliver automatic conveyance control, and supported dynamic scheduling demands including order adjustment, cancellation and emergency insertion.

SKILLS

Languages Chinese (Native), English (Proficient)

Computer Python (Proficient), Java (Proficient), C++ (3 yrs), C (Proficient), Photoshop, AutoCAD, Corel Draw, Microsoft Office

Hobbies Traveling, Tennis, Badminton, Calligraphy, Piano (6 years of stage performances)